

Biostatistics handover

Phase III trial · 142 outputs · CDISC submission package · SAP signed before lock · independently double programmed

This is a redacted sample. It shows the structure and level of detail of a real deliverable. Sponsor, compound and indication have been removed or changed. Ask us and we will send one built around a study like yours.

1 The statistical analysis plan

Signed and dated eleven days before database lock. That date is what makes every analysis below pre-specified, and it is the first thing an agency statistician checks.

Element	As specified
Estimand	Treatment effect regardless of adherence; intercurrent events handled by a treatment policy strategy, per ICH E9(R1)
Primary analysis	Cox proportional hazards, stratified by region and disease stage
Populations	Intention to treat as primary; per protocol and safety populations defined separately
Missing data	Multiple imputation, with a pre-specified tipping-point sensitivity analysis
Multiplicity	Hierarchical testing across three secondary endpoints, order fixed in advance
Interim analysis	One efficacy look at 60% of events, O'Brien-Fleming boundary
Sample size	HR 0.72, 90% power, two-sided 5%, 12% drop-out — assumptions sourced to two cited trials
Proportional hazards	Assessed by scaled Schoenfeld residuals; restricted mean survival time pre-specified as the fallback

Why the fallback matters. A Cox model assumes the hazard ratio is constant over time. When it is not — which happens often with immunotherapies, and with any treatment whose effect takes months to appear — the reported hazard ratio is an average of something that changed, and a reviewer will say so. Naming the alternative in advance turns that from a crisis into a paragraph.

2 Datasets

Deliverable	Detail
SDTM domains	19 domains mapped from the EDC extract, controlled terminology applied, annotated CRF supplied
ADaM datasets	ADSL, ADTTE, ADAE and six analysis datasets, every derived value traceable to its SDTM source
Define.xml	Generated from the datasets rather than written alongside them, so the two cannot drift apart

Deliverable	Detail
Reviewer's guides	Study data and analysis data guides, with every validation warning explained
Conformance	Pinnacle 21: zero errors, four warnings, each documented

Raised with the sponsor rather than resolved quietly

- Fourteen participants had a visit date preceding their consent date. Eleven were transcription errors corrected through the query process; three were genuine protocol deviations and are listed as such.
- The stratification factor recorded at randomisation disagreed with the case report form for nine participants. Analysed as randomised, with a pre-specified sensitivity analysis using the corrected strata, and both results reported.
- One site contributed 18% of participants and had a markedly different event rate. A leave-one-site-out sensitivity analysis was added, agreed in writing before it was run.

3 Validation

Every one of the 142 tables, listings and figures was produced twice: independently, by a second programmer working from the analysis plan rather than from the first programmer's code, and reconciled before release.

Check	Result
Outputs double programmed	142 of 142
Discrepancies found	7
In the validating programmer's work	5
In the primary programmer's work	2
Unresolved at release	0

The two that mattered. One of the primary programmer's errors would have mis-stated an adverse event denominator, understating an incidence rate in a safety table. It was found because someone else produced the same table from the plan and got a different number. This is the entire argument for double programming, and it is the line item sponsors most often ask to cut when a timeline slips.

Deviation from the plan, declared

One pre-specified subgroup contained too few events to fit the planned stratified model. It is reported as not estimable rather than refitted into something that would produce a figure, and the deviation is listed in the clinical study report alongside the reason. A subgroup quietly refitted until it yields a number is the kind of thing that is invisible in a paper and obvious in an audit.

4 What is handed over

- The signed statistical analysis plan, with a written log of every deviation from it.
- SDTM and ADaM datasets as XPT and CSV, with Define.xml and the annotated CRF.
- 142 tables, listings and figures as RTF and PDF, built to the sponsor's shell specifications.

- Kaplan-Meier, forest and CONSORT flow figures as vector files at submission resolution.
- All SAS and R programmes, commented and version controlled, with their run logs and software versions.
- The study data and analysis data reviewer's guides.
- The validation record: the double-programming log, the seven discrepancies and their resolutions.

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